CSC649 — SPECIAL TOPICS IN COMPUTER SCIENCE

Assignment 1: Problem Formulation, Requirements & System Diagrams

Project Focus: Intelligent Canteen Diet-Matching Decision Support System

1. Part A: Problem Formulation

1.1 Problem Statement (Part A1)

University students and staff sometimes wanted to follow specific health, fitness, or medical diets struggle to make informed food choices at campus canteens due to a complete lack of transparent nutritional information. Cafeteria stalls offer diverse local dishes without displaying calorie counts, macronutrient breakdowns (protein, carbohydrates, fats), or ingredient details, leaving diners to estimate their nutritional intake blindly. As a concrete example, a student trying to maintain a calorie deficit or high-protein diet for fitness goals cannot accurately track meals when choosing mixed rice, often resulting in poor dietary adherence and frustration. This systemic information gap makes consistent healthy eating on campus nearly impossible, negatively impacting student wellness and academic energy levels.

1.2 Users & Scope

Main Users & Stakeholders:

- **Students & Campus Staff (End Users):** Individuals seeking personalized meal recommendations matching daily calorie and macro goals, allergies, or religious dietary restrictions.
- **Canteen Vendors & Stall Operators:** Providing daily menu offerings, ingredient components, pricing, and availability updates.
- **Campus Health & Wellness Administration:** Institutional stakeholders interested in promoting healthier campus dining habits and tracking aggregated nutrition trends.

Project Scope Boundaries:

In-Scope Features	Out-of-Scope Elements
• Capturing daily canteen menu items and mapping them to nutritional reference data.	• Physical preparation, cooking, or kitchen management by vendors.

• Intelligent fuzzy rule engine to rank dishes based on user constraints (calories, protein, budget, halal). • Explanation generation justifying why specific meals match user goals. • User profile & macro target management dashboard.	• Real-time automated visual plate scanning via computer vision (future phase). • Payment gateway integration or online food delivery dispatch.

2. Part B: System Requirements & User Stories

2.1 Functional Requirements

Functional Requirements Statement	Verification Method
The system shall allow users to set and update their daily target calories, macronutrient targets (protein, carbs, fats), and dietary restrictions (vegetarian, nut allergy).	Test Case / UI Test
The system shall allow canteen vendors to input and update daily menu items, dish ingredients, estimated portion sizes, and pricing per stall.	Data Entry Test
The system shall process remaining daily macros using a Fuzzy Logic Engine to evaluate and rank optimal dish combinations from available stalls.	Algorithm Test
The system shall display recommended meal options accompanied by a natural language decision rationale	UI Verification

2.2 Non-Functional Requirements (Part B2)

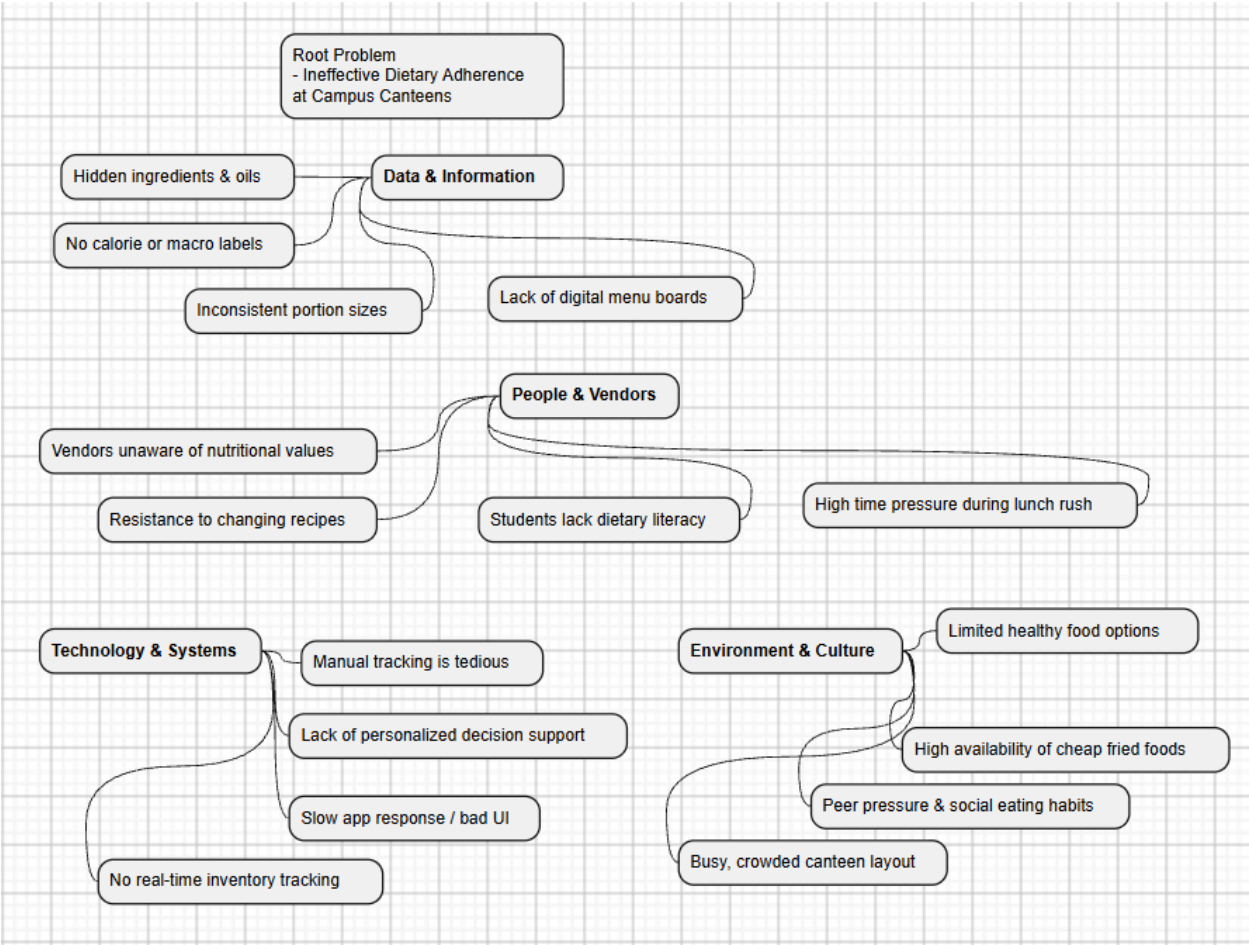
Performance: The system shall generate personalized meal recommendations within 2.0 seconds of request under peak server load.

Usability: The user interface shall enable a student to view and select a recommended meal option in no more than 3 screen taps from launch.

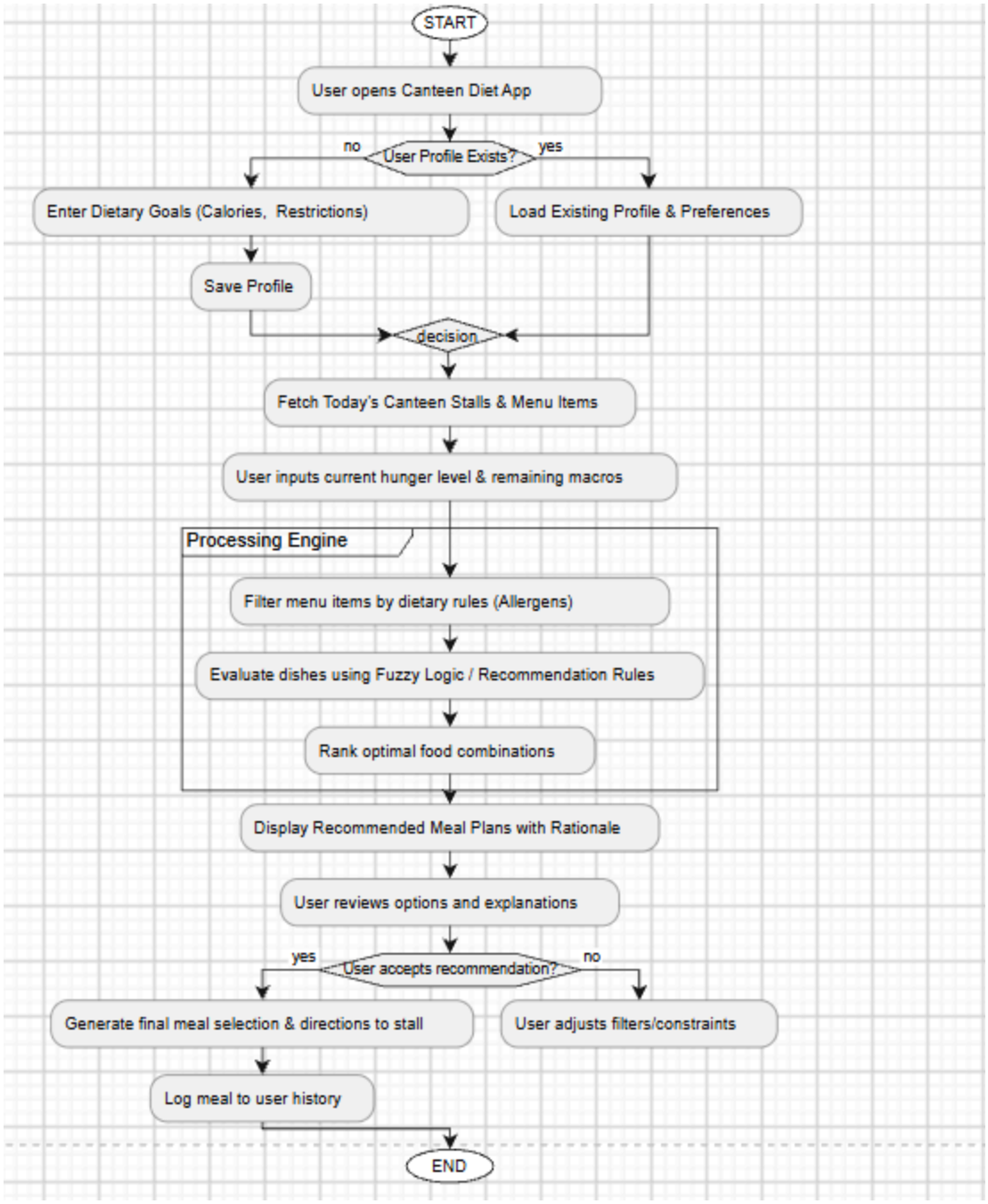
Security & Privacy: The system shall encrypt all stored user personal health profiles using AES-256 standard and adhere to data protection policies.

3. System Modeling & Architectural Diagrams

3.1 Cause-and-Effect Analysis: Fishbone Diagram



3.2 User Workflow: Activity Diagram


START
User opens Canteen Diet App
User Profile Exists?
no
yes
Enter Dietary Goals (Calories, Restrictions)
Load Existing Profile & Preferences
Save Profile
decision
Fetch Today's Canteen Stalls & Menu Items
User inputs current hunger level & remaining macros
Processing Engine
Filter menu items by dietary rules (Allergens)
Evaluate dishes using Fuzzy Logic / Recommendation Rules
Rank optimal food combinations
Display Recommended Meal Plans with Rationale
User reviews options and explanations
User accepts recommendation?
yes
no
Generate final meal selection & directions to stall
User adjusts filters/constraints
Log meal to user history
END